Topic: Week 7 – SQL Query Implementation for E-Commerce Database

1. Display All Customer Details

```
SQL> SELECT * FROM Customer;

CUSTOMER_ID CUSTOMER_NAME
----------- ------------------------------------------------------------
EMAIL                                                        PHONE
------------------------------------------------------------ ----------
       1001 Arun
arun@gmail.com                                               9876543210

       1002 Priya
priya@gmail.com                                              9876543211

       1003 Kumar
kumar@gmail.com                                              9876543212


CUSTOMER_ID CUSTOMER_NAME
----------- ------------------------------------------------------------
EMAIL                                                        PHONE
------------------------------------------------------------ ----------
       1004 Divya
divya@gmail.com                                              9876543213

       1005 Rahul
rahul@gmail.com                                              9876543214
```

2. Display all available products

```
SQL> SELECT * FROM Product;

PRODUCT_ID PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
---------- -------------------- ------- ---------- --------------
P1001      Laptops              Cat0001      67500              2
P1002      Mobile               Cat0001      25000             10
P1003      TShirt               Cat0002        800             20
P1004      Shoes                Cat0003       1500             15
```

3.Retrieve product names and prices only:

```
SQL> SELECT Product_Name, Price FROM Product;

PRODUCT_NAME             PRICE
-------------------- ----------
Laptops                   67500
Mobile                    25000
TShirt                      800
Shoes                      1500
```

4.Display All Orders Placed by Customers

```
SQL> SELECT * FROM Orders;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
      1001 27-SEP-26        67500 Pending           1001
      1002 27-SEP-26        25000 Shipped           1002
      1003 27-SEP-26         1500 Delivered         1003
      1004 27-SEP-26         4200 Pending           1004
      1005 27-SEP-26         8000 Delivered         1005
```

5.Retrieve Payment Details

```
SQL> SELECT * FROM Payment;
PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_ST
---------- ---------- --------- -------------------- ----------
TRANSACTION_AMOUNT
------------------
         1       1001 27-SEP-26 UPI                  Success
             67500

         2       1002 27-SEP-26 Card                 Success
             25000

         3       1003 28-SEP-26 Cash                 Success
              1500

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_ST
---------- ---------- --------- -------------------- ----------
TRANSACTION_AMOUNT
------------------
         4       1004 28-SEP-26 Net Banking          Pending
              4200

         5       1005 29-SEP-26 UPI                  Success
              8000

         6       1001 29-SEP-26 Card                 Failed
              1000

PAYMENT_ID   ORDER_ID PAYMENT_D PAYMENT_MODE         PAYMENT_ST
---------- ---------- --------- -------------------- ----------
TRANSACTION_AMOUNT
------------------
         7       1002 30-SEP-26 UPI                  Pending
              5000

         8       1003 30-SEP-26 Net Banking          Failed
              2000

8 rows selected.
```

2.Apply Filtering Conditions Using WHERE

1.Find Products with Price Greater Than ₹5000

```
SQL> SELECT * FROM Product WHERE Price > 5000;

PRODUCT_ID PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
---------- -------------------- ------- ---------- --------------
P1001      Laptops              Cat0001      67500              2
P1002      Mobile               Cat0001      25000             10
```

2.Display Products Available in Stock

```
SQL> SELECT * FROM Inventory WHERE Stock_Status = 'Available';

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
------- ------- ------- -------------- --------------- ---------
I001    P1001   S001                20 Available       20-SEP-26
I002    P1002   S002                15 Available       20-SEP-26
I003    P1003   S003                10 Available       20-SEP-26
```

3.Display Completed Orders

```
SQL> SELECT * FROM Orders WHERE Order_Status = 'Delivered';

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
      1003 27-SEP-26         1500 Delivered         1003
      1005 27-SEP-26         8000 Delivered         1005
```

Find Products with Rating Above 4

```
SQL> SELECT P.Product_Name, ROUND(AVG(R.Rating), 2) AS Average_Rating FROM Product P JOIN Review R ON P.Product_ID = R.Product_ID GROUP BY
 P.Product_ID, P.Product_Name HAVING AVG(R.Rating) > 4 ORDER BY Average_Rating DESC;

PRODUCT_NAME       AVERAGE_RATING
------------------ --------------
TShirt                          5
Laptops                         5
```

3.SORT DATA USING ORDER BY

1.Display Products from Lowest to Highest Price

```
SQL> SELECT * FROM Product ORDER BY Price ASC;

PRODUCT_ID PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
---------- -------------------- ------- ---------- --------------
P1003      TShirt               Cat0002        800             20
P1004      Shoes                Cat0003       1500             15
P1002      Mobile               Cat0001      25000             10
P1001      Laptops              Cat0001      67500              2
```

2.Display Customers Alphabetically

```
SQL> SELECT * FROM Customer ORDER BY Customer_Name ASC;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                    PHONE
-------------------------------------------------- ----------
       1001 Arun
arun@gmail.com                                     9876543210

       1004 Divya
divya@gmail.com                                    9876543213

       1003 Kumar
kumar@gmail.com                                    9876543212


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                    PHONE
-------------------------------------------------- ----------
       1002 Priya
priya@gmail.com                                    9876543211

       1005 Rahul
rahul@gmail.com                                    9876543214
```

3.Find top expensive products

```
SQL> SELECT * FROM Product ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
---------- -------------------- ------- ---------- --------------
P1001      Laptops              Cat0001      67500              2
P1002      Mobile               Cat0001      25000             10
P1004      Shoes                Cat0003       1500             15
P1003      TShirt               Cat0002        800             20
```

4.Display Latest Orders First

```
SQL> SELECT * FROM Orders ORDER BY Order_Date DESC;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
      1005 27-SEP-26         8000 Delivered          1005
      1004 27-SEP-26         4200 Pending            1004
      1003 27-SEP-26         1500 Delivered          1003
      1002 27-SEP-26        25000 Shipped            1002
      1001 27-SEP-26        67500 Pending            1001
```

4.Retrieve unique values using DISTINCT

1.Display unique product categories

```
SQL> SELECT DISTINCT Category_ID FROM Product;

CATEGOR
-------
Cat0001
Cat0002
Cat0003
```

2.Find Different Payment Methods Used by Customers

```
SQL> SELECT DISTINCT Payment_Mode AS Payment_Method FROM Payment;

PAYMENT_METHOD
--------------------
UPI
Card
Cash
Net Banking
```

5.Search Products Based on Conditions

1.Find Products Within a Specific Price Range Using BETWEEN – Display products priced between ₹20,000 and ₹40,000.

```
SQL> SELECT * FROM Product WHERE Price BETWEEN 20000 AND 40000;

PRODUCT_ID PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
---------- -------------------- ------- ---------- --------------
P1002      Mobile               Cat0001      25000             10
```

2.Display Products Belonging to a Particular Category

```
SQL> SELECT * FROM Product WHERE Category_ID = 'Cat0001';

PRODUCT_ID PRODUCT_NAME         CATEGOR      PRICE STOCK_QUANTITY
---------- -------------------- ------- ---------- --------------
P1001      Laptops              Cat0001      67500              2
P1002      Mobile               Cat0001      25000             10
```

3.Find Products Currently Available in Inventory

```
SQL> SELECT * FROM Inventory WHERE Stock_Quantity > 0;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATUS    LAST_UPDA
------- ------- ------- -------------- --------------- ---------
I001    P1001   S001                20 Available       20-SEP-26
I002    P1002   S002                15 Available       20-SEP-26
I003    P1003   S003                10 Available       20-SEP-26
```

4.Search Products Using Product Name

```
SQL> SELECT * FROM Product WHERE Product_Name = 'Mobile';

PRODUCT_ID PRODUCT_NAME          CATEGOR      PRICE STOCK_QUANTITY
---------- --------------------- ------- ---------- --------------
P1002      Mobile                Cat0001      25000             10
```

5.Find Low-Stock Products

```
SQL> SELECT * FROM Product WHERE Stock_Quantity < 5;

PRODUCT_ID PRODUCT_NAME          CATEGOR      PRICE STOCK_QUANTITY
---------- --------------------- ------- ---------- --------------
P1001      Laptops               Cat0001      67500              2
```

6.Retrieve customer and product information

1.Display Customer Details with Their Orders

```
SQL> SELECT C.Customer_ID, C.Customer_Name, O.Order_ID, O.Order_Date, O.Total_Amount, O.Order_Status FROM Customer C JOIN Orders O ON C.Cu
stomer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ---------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------
       1001 Arun                                            1001
27-SEP-26        67500 Pending

       1002 Priya                                           1002
27-SEP-26        25000 Shipped

       1003 Kumar                                           1003
27-SEP-26         1500 Delivered


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ---------
ORDER_DAT TOTAL_AMOUNT ORDER_STAT
--------- ------------ ----------
       1004 Divya                                           1004
27-SEP-26         4200 Pending

       1005 Rahul                                           1005
27-SEP-26         8000 Delivered
```

2.Display product details with category information

```
SQL> SELECT P.Product_ID, P.Product_Name, C.Category_Name, P.Price, P.Stock_Quantity FROM Product P JOIN Category C ON P.Category_ID = C.C
ategory_ID;

PRODUCT_ID PRODUCT_NAME         CATEGORY_NAME                    PRICE
---------- -------------------- -------------------------------- ----------
STOCK_QUANTITY
--------------
P1001      Laptops              Electronics                      67500
             2

P1002      Mobile               Electronics                      25000
            10

P1003      TShirt               Clothing                           800
            20


PRODUCT_ID PRODUCT_NAME         CATEGORY_NAME                    PRICE
---------- -------------------- -------------------------------- ----------
STOCK_QUANTITY
--------------
P1004      Shoes                Footwear                          1500
            15
```

3.Find customers who purchased a specific product

```
SQL> SELECT DISTINCT C.Customer_ID, C.Customer_Name, O.Order_ID, OD.Product_ID FROM Customer C JOIN Orders O ON C.Customer_ID = O.Customer
_ID JOIN Order_Details OD ON O.Order_ID = OD.Order_ID WHERE OD.Product_ID = 'P1001';

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ---------
PRODUCT
-------
       1001 Arun                                                    1001
P1001
```

4.Display products purchased by each customer

```
SQL> SELECT C.Customer_Name, O.Order_ID, P.Product_Name, OD.Quantity FROM Customer C JOIN Orders O ON C.Customer_ID = O.Customer_ID JOIN O
rder_Details OD ON O.Order_ID = OD.Order_ID JOIN Product P ON OD.Product_ID = P.Product_ID;

CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ---------
PRODUCT_NAME          QUANTITY
--------------------- ---------
Arun                                                    1001
Laptops                       1

Priya                                                   1002
Mobile                        1

Kumar                                                   1003
TShirt                        2


CUSTOMER_NAME                                      ORDER_ID
-------------------------------------------------- ---------
PRODUCT_NAME          QUANTITY
--------------------- ---------
Divya                                                   1004
Shoes                         2
```

7.Apply mutiple filtering conditions

1.Find electronics products costing more than rs 10000

```
SQL> SELECT P.Product_ID, P.Product_Name, P.Category_ID, P.Price FROM Product P WHERE P.Category_ID = 'Cat0001' AND P.Price > 10000 ORDER
BY P.Price DESC;

PRODUCT_ID PRODUCT_NAME        CATEGOR     PRICE
---------- -------------------- ------ ---------
P1001      Laptops              Cat0001     67500
P1002      Mobile               Cat0001     25000
```

2.Search Products Containing the Word “Mobile”

```
SQL> SELECT Product_ID, Product_Name, Price, Stock_Quantity FROM Product WHERE UPPER(Product_Name) LIKE '%MOBILE%';

PRODUCT_ID PRODUCT_NAME              PRICE STOCK_QUANTITY
---------- -------------------- ---------- --------------
P1002      Mobile                    25000             10
```

3.Find Orders Placed Within a Specific Date Range

```
SQL> SELECT * FROM Orders WHERE Order_Date >= DATE '2026-09-27' AND Order_Date < DATE '2026-09-28' ORDER BY Order_Date;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STAT CUSTOMER_ID
---------- --------- ------------ ---------- -----------
      1001 27-SEP-26        67500 Pending           1001
      1002 27-SEP-26        25000 Shipped           1002
      1003 27-SEP-26         1500 Delivered         1003
      1004 27-SEP-26         4200 Pending           1004
      1005 27-SEP-26         8000 Delivered         1005
```

8.Generate basic business reports Report

1: product availability report

Display :

1.product name 2.price 3.stock quantity 4.availability status

```
SQL> SELECT Product_Name, Price, Stock_Quantity, CASE WHEN Stock_Quantity > 0 THEN 'Available' ELSE 'Out of Stock' END AS Availability_Sta
tus FROM Product;

PRODUCT_NAME              PRICE STOCK_QUANTITY AVAILABILITY
-------------------- ---------- -------------- ------------
Laptops                   67500              2 Available
Mobile                    25000             10 Available
TShirt                      800             20 Available
Shoes                      1500             15 Available
```

Report 2: customer report

Total number of customers

```
SQL> SELECT COUNT(*) AS Total_Customers FROM Customer;

TOTAL_CUSTOMERS
---------------
              5
```

New customer registrations

```
SQL> SELECT Customer_ID, Customer_Name, Email FROM Customer ORDER BY Customer_ID DESC;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------
       1005 Rahul
rahul@gmail.com

       1004 Divya
divya@gmail.com

       1003 Kumar
kumar@gmail.com


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL
--------------------------------------------------
       1002 Priya
priya@gmail.com

       1001 Arun
arun@gmail.com
```

Report 3: order report

display: Total orders

```
SQL> SELECT COUNT(*) AS Total_Orders FROM Orders;

TOTAL_ORDERS
------------
           5
```

Completed orders

```
SQL> SELECT COUNT(*) AS Completed_Orders FROM Orders WHERE Order_Status = 'Delivered';

COMPLETED_ORDERS
----------------
               2
```

Pending orders

```
SQL> SELECT COUNT(*) AS Pending_Orders FROM Orders WHERE Order_Status = 'Pending';

PENDING_ORDERS
--------------
             2
```

Cancelled orders

```
SQL> SELECT COUNT(*) AS Cancelled_Orders FROM Orders WHERE Order_Status = 'Cancelled';

CANCELLED_ORDERS
----------------
               0
```

Report 4: product performance report

Higest priced products

```
SQL> SELECT Product_ID, Product_Name, Price FROM Product ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME              PRICE
---------- -------------------- ----------
P1001      Laptops                   67500
P1002      Mobile                    25000
P1004      Shoes                      1500
P1003      TShirt                      800
```

Most reviewed products

```
SQL> SELECT P.Product_ID, P.Product_Name, COUNT(R.Product_ID) AS Total_Reviews FROM Product P JOIN Review R ON P.Product_ID = R.Product_ID
 GROUP BY P.Product_ID, P.Product_Name ORDER BY Total_Reviews DESC;

PRODUCT_ID PRODUCT_NAME         TOTAL_REVIEWS
---------- -------------------- -------------
P1001      Laptops                          2
P1003      TShirt                           1
P1004      Shoes                            1
P1002      Mobile                           1
```

Available products

```
SQL> SELECT Product_ID, Product_Name, Stock_Quantity FROM Product WHERE Stock_Quantity > 0;

PRODUCT_ID PRODUCT_NAME         STOCK_QUANTITY
---------- -------------------- --------------
P1001      Laptops                           2
P1002      Mobile                           10
P1003      TShirt                           20
P1004      Shoes                            15
```